INCH-POUND

MIL-PRF-38715B
15 October 1996
SUPERSEDING
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20 April 1971

PERFORMANCE SPECIFICATION

INDICATOR, COMBUSTIBLE GAS AND OXYGEN

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope. This specification covers two types of indicators for measuring combustible gas and oxygen.

1.2 Classification. Indicators covered by this specification shall be of the following types as specified (see 6.2).

Type I - Indicator, combustible gas and oxygen, without calibrator.
Type II - Indicator, combustible gas and oxygen, with calibrator.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/TILDD, 485 Quentin Roosevelt Rd. Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 And JP-5
MIL-C-38736	Cleaning Compound, Solvent Mixture
MIL-T-83133	Jet Fuel, JP-8
MIL-P-87107	Missile Fuel, JP-10

STANDARDS

MILITARY

MIL-STD-810	Environmental Test Methods

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

National Fire Protection Association

NFPA 70	National Electric Code

(Application for copies should be addresses to the National Fire Protection Association, Batterymarch Park, Quincy, MA, 02269-9990)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified (see 6.2), a sample(s) shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The contractor shall select the materials, but the materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.1 Metals. Metals shall be of the corrosion-resistant type and/or be suitably treated to resist corrosion during normal service life. Materials contained within hermetically sealed enclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment, anodizing of aluminum alloy parts, and nickel plating of steel parts are not applicable for parts within hermetically sealed instruments. Stainless steel parts, steel parts that have been nickel plated, and aluminum alloy parts that have been anodized, shall be further protected against corrosion if necessary.

3.2.2 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.2.3 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The usage of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.4 Explosive resistance. The indicator shall be constructed intrinsically safe for Class I, Division 1, Group A, B, C and D locations (as defined by National Fire Protection Association Code 70 Article 500).

3.3 Physical characteristics.

3.3.1 Weight. The total weight of the indicator, both types, including batteries shall not exceed 20 pounds.

3.3.2 Dimensions. The indicator shall not be larger than 2 feet in any dimension.

3.3.3 Controls. The indicator shall be provided with a control(s) for obtaining the initial zero setting of the indicator and shall have a control(s) for changing the sensitivity (ranges) of the indicator. When set, controls shall not be accidentally disturbed. Controls must be usable with cold weather gear gloves.

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3.3.4 Display. The indicator shall show the chosen range of Lower Explosive Limit (LEL) on a display that is readable in full sunlight from a distance of no less than 70 inches. The display shall show measured LEL percentages for the chosen gas and range on a scale.

3.3.5 Scale ranges The scales shall be either digital or analog. The range of the combustible gas scale shall be 0 to 50 percent(%) LEL and 0 to 5% LEL. It shall consist of 50 scale divisions with figures at the 0, 10, 20, 30, 40, and 50% graduations. The 0 to 5% LEL scale shall use the same divisions as the 0 to 50% LEL scale with 1% corresponding to the 10% figure on the 0-50% scale, 2% corresponding to 20%, etc. The range of the oxygen scale shall be 0-25%. It shall consist of 25 equal scale divisions with figures at 0, 5, 10, 15, 20 and 25% graduations. The scale division requirement applies only when analog scales are used. A switch shall be provided to change the scale of the dual range combustible gas scale from 0-50% LEL to the 0-5% LEL scale, or the indicator shall automatically switch scales. For digital indicators, the display shall be rounded to the nearest 0.1% with .05% rounding up.

3.3.6 Construction. The indicator shall be capable of meeting all of the performance requirements after exposure to normal handling, to include dropping from no more then 3 feet.

3.4 Performance characteristics

3.4.1 Gas detection. The indicator shall detect explosive vapor or gases associated with fuel oils or gasoline (including leaded gasoline), alcohol vapors, and hydrogen in mixtures with air. For each of the following gases and/or vapors, the indicator shall automatically (within 5 seconds) provide the true percent LEL after the user sets the gas selector switch to the appropriate gas.

a. Jet, rocket and reciprocating engine fuels:

(1) JP-4 (MIL-T-5624)

(2) JP-5 (MIL-T-5624)

(3) JP-8 (MIL-T-83133)

(4) PF-1

(5) JP-10 (MIL-P-87107)

(6) 4-Part Cleaner (MIL-C-38736)

(7) Aviation gasoline 100-130 octane

b. Ethyl alcohol

c. Hydrogen

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d. Benzene

e. Methane

f. Hexane

g. Pentane

h. Heptane

3.4.2 Stability. The indicator shall display a stable reading of the selected gas with a variation of no more or less than 10% of full scale.

3.4.3 Ranges of detection. The combustible gas scale shall have a range of 0-5% and 0-50% Lower Explosive Limit (LEL) for the selected gas in any ambient temperature from -18 to 52 degrees Centigrade (°C). The oxygen scale shall have a range of 0-25% and shall be temperature compensated to measure oxygen in the range of -18°C to 52°C.

3.4.4 Accuracy. On the combustible scales, the indicated concentration expressed in percent of the LEL of any gas or vapor, shall not be less than the true concentration or exceed the true concentration by more than 10% of full scale for both the 0-5% LEL and 0-50% LEL scales. On the oxygen scale, the difference between indicated concentration and the true concentration shall not be greater than plus or minus 1.5% oxygen when tested in the range compensated for temperature.

3.4.5 Alarm. The indicator shall include audible and visual alarm signals. The set point for the alarm signals shall be adjustable over the full scale ranges of the instrument.

3.4.6 Calibration. The indicator calibration reference shall be a standard gas (recommended by the manufacturer and approved by the government). Calibration shall be achieved by adjusting an external or internal potentiometer(s) appropriately for the calibrating gas or gases.

3.4.7 Power supply. The indicator shall run on rechargeable batteries. The batteries shall remain fully operational for no less than 8 continuous hours. The batteries shall be commercially available from a minimum of two sources (may include the item manufacturer). The batteries shall be user replaceable.

3.5 Environmental conditions.

3.5.1 Temperature. The indicator shall be capable of satisfying all of the performance requirements after exposure to a operating temperature range of -18°C to 52°C.

3.5.2 Temperature shock. The indicator shall be capable of satisfying all of the performance requirements after exposure to a temperature shock of -18 to 21°C.

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3.5.3 Rain. The indicator shall be capable of satisfying all of the performance requirements during and after exposure to rain.

3.5.4 Humidity. The indicator shall be capable of satisfying all of the performance requirements after exposure to high humidity conditions.

3.5.5 Salt. The indicator shall be capable of satisfying all of the performance requirements after exposure to a salt air environment.

3.5.6 Vibration. The indicator shall be capable of satisfying all of the performance requirements after exposure to vibration during transportation and mobilization.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete indicator assembly when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1 - 4.5.16 done in the order stated. When determined necessary by the government, a separate indicator shall be used for the fungus test of 4.5.11.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.1-4.5.5.

4.4 Examination. Each indicator shall be examined for compliance with the requirements. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability.

4.4.1 Weight. The indicator shall be examined for compliance with the weight restriction of 20 pounds in accordance with 3.3.1.

4.4.2 Dimensions. The indicator shall be examined for compliance with the dimensional restriction of 2 feet in accordance with 3.3.2.

4.4.3 Controls. The indicator shall be visually examined for compliance with 3.3.3.

4.4.4 Display. The indicator shall be visually examined for compliance with 3.3.4.

4.4.5 Scale ranges. The indicator shall be visually examined for compliance with 3.3.5.

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4.5 Methods of inspection.

4.5.1 Response. The indicator shall respond to a 50 percent LEL methane-air mixture within 5 seconds at standard temperature.

4.5.2 Scale accuracy. The indicator shall be tested with 25% and 50% LEL mixtures (gases of para.3.4.1) in air as well as with a 15% oxygen in nitrogen mixture. The indicated concentration shall not be less than the true concentration or exceed the true by more than 10 percent of full scale. The oxygen detector shall be tested for compliance with para 3.4.4. Finally, at the discretion of the government inspector, the contractor may be required to demonstrate the satisfactory operation of any indication.

4.5.3 Alarm. The alarm shall be set at a chosen percent LEL below that used in 4.5.2 and then tested for compliance with requirement 3.4.5 during a test for scale accuracy in 4.5.2.

4.5.4 Calibration. The indicator shall be calibrated with a standard gas and tested for scale accuracy (4.5.2).

4.5.5 Stability. On each range, the indicator shall be operated on fresh air for 15 minutes, and "zeroed" by means of the proper controls. On each range, the indicator shall then be operated on a methane-air mixture of known concentration and identity for 3 consecutive periods of 5 minutes each, while being quickly and accurately "zeroed" again on fresh air before each period of 5 minutes. The variation in reading shall not be lower than the true concentration nor exceed the true concentration by more than 10 percent of full scale.

4.5.6 High temperature. The indicator shall be mounted in its normal operating position in a chamber. The test shall be performed in accordance with Method 501.3, Procedure II of MIL-STD-810, Operation. The high end temperature shall be set at 52°C. The indicator shall then be tested for scale accuracy (4.5.2).

4.5.7 Low temperature. The indicator shall be mounted in its normal operating position in a chamber. The test shall be performed in accordance with Method 502.3, Procedures II and III of MIL-STD-810, Operation and Manipulation. The low end temperature shall -18°C. The indicator shall then be tested for scale accuracy (4.5.2).

4.5.8 Temperature shock. The indicator shall be mounted in its normal operating position in a chamber. The test shall be performed in accordance with Method 503.3, Procedure III of MIL-STD-810, Ground Transfer. The low end temperature used shall be -18°C. The indicator shall then be tested for scale accuracy (4.5.2).

4.5.9 Rain. The indicator shall be mounted for one hour with the dial/display in a vertical plane and one hour with the dial/display in a horizontal plane (face upward) in a rain test chamber. The external connections shall be made to the indicator in such a manner as to simulate installed conditions. The test shall be performed in accordance with Method 506.3, Procedures I and II of

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MIL-STD-810, Blowing Rain and Drip. After the test, the indicator shall be examined to ascertain that no water has entered the indicator

4.5.10 Humidity. The indicator shall be mounted in its normal operating position in a chamber. The test shall be performed in accordance with Method 507, Procedure I of MIL-STD-810, Natural. After the test, the indicator will be examined to ascertain that corrosion is not present and subsequent operation is not adversely affected.

4.5.11 Fungus. The indicator, including applicable external connections, shall be subjected to the action of fungi in accordance with Method 508.4, Procedure I of MIL-STD-810. After the test, the indicator shall be examined to ascertain that no fungus growth has occurred. At the option of the inspector, tests in addition to the above may be made to determine that there has been no deterioration which would adversely affect subsequent operation.

4.5.12 Salt. The indicator shall be subjected to salt spray in accordance with Method 509.3, Procedure I of MIL-STD-810. Evidence of corrosion or discoloration of any part of the indicator shall constitute failure of the indicator to pass this test.

4.5.13 Vibration. The indicator shall be subjected to vibration in accordance with Method 514.4, Procedure I of MIL-STD-810, Basic Transportation. There shall be no looseness in the mechanism nor damage to any part of the indicator, as a result of the vibration. The indicator shall be securely mounted on a vibration table during this test. The indicator shall then be tested for scale accuracy (4.5.2).

4.5.14 Endurance. The indicator shall be operated for 24 hours at 25% of the LEL with methane (or a government approved substitute) and air. The indicator shall be serviceable for a continuous period of at least 8 hours. The indicator shall be serviceable and accurate within the limits specified for scale accuracy in 4.5.2, at the end of the 24 hour period.

4.5.15 Shock. The indicator shall be subjected to shock in accordance with Method 516.4, Procedures I and III of MIL-STD-810, Functional Shock and Fragility. The indicator shall then be tested for scale accuracy (4.5.2).

4.5.16 Explosive resistance. Documentation shall be provided for each type indicator that proves it is intrinsically safe for Class I, Division 1, Group A, B, C and D locations (as defined by National Fire Protection Association Code 70 Article 500).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military

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Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory)

6.1 Intended use. This portable combustible gas indicator covered by this specification is intended for testing air in fuel tanks and other enclosed spaces for the presence of dangerous concentrations of combustible gases or vapors.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Location and conditions for preproduction testing.

c. Applicable levels of preservation, packaging, and packing.

d. Quantity of unit packages to be included in each intermediate package.

e. Quantity of calibrators to be procured.

f. Type (Type I or Type II, see 1.2).

g. First Article Requirement

6.3 Subject term (key word) listing.

Aviation gasoline 100-130 octane	Benzene
Combustible gas indicator	Ethyl alcohol
Heptane	Hexane
Hydrogen	JP-4
JP-5	JP-8
JP-10	Methane
Oxygen indicator	PF-1
Pentane	

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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3. DOCUMENT
INDICATOR, COMBUSTIBLE GAS AND OXYGEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as*

5. REASON FOR

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area*
SA-ALC/TILDD	(1) (210) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 Quentin Roosevelt Rd Kelly AFB, TX 78241-6416	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466

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